Practice workbook · Read and write text files line by line

The program opens a text file to read lines (strip, split on commas, cast, check or total) and/or writes records to a file (from input or from an array, comma-separated, one record per line), then closes the file.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-CODE-08

1. Question 5 · part c · 8 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part c only.

5 Programmers refine algorithms, use searches and read data from files.

(a) **Figure 6** shows an algorithm that determines whether the animal name inputted by the user is found.

```
1  SET a1 TO 'alpaca'
2  SET a2 TO 'bear'
3  SET a3 TO 'camel'
4  SET a4 TO 'deer'
5
6  SEND ('Enter an animal: ') TO DISPLAY
7  RECEIVE target FROM (STRING) KEYBOARD
8
9  IF (a1 = target) THEN
10     SEND ('Found') TO DISPLAY
11 ELSE
12     IF (a2 = target) THEN
13         SEND ('Found') TO DISPLAY
14     ELSE
15         IF (a3 = target) THEN
16             SEND ('Found') TO DISPLAY
17         ELSE
18             IF (a4 = target) THEN
19                 SEND ('Found') TO DISPLAY
20             END IF
21         END IF
22     END IF
23 END IF
```

Figure 6

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1. Question 5 · part c · 8 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part c only.

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Explain **one** improvement to this algorithm that will reduce the number of variables required and will enable it to work with any number of animals.

(2)

(b) Binary search is a divide and conquer algorithm.

Here is a list of data.

10	12	13	14	15	16	17	18	19	20

A binary search is used to check whether the number 11 is in the list.

The midpoint is calculated by adding the high index to the low index and **integer dividing** by 2

Complete the table to show the values for the high index, the low index and the midpoint on each pass of a binary search.

You may not need to use all the rows in the table.

(4)

high index	low index	midpoint

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1. Question 5 · part c · 8 marks

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Worked-example retry · Attempt part c only.

(c) **Figure 7** shows the **Sales.txt** file. It stores sales information.

```
264,140,120,284,192
420,377,435,376,392
619,589,606,586,600
799,811,788,814,788
982,1007,1013,989,1009
```

Figure 7

A program is required to calculate and display:

- a subtotal for each line of sales
- a grand total for all the sales in the file.

Figure 8 shows the intended output from the program.

```
1000
2000
3000
4000
5000
Grand total: 15000
```

Figure 8

Open **Q05c** in the code editor.

Amend the code to produce the intended output.

You must use the structure and variables given in **Q05c** to complete the program.

Do not add any further functionality.

Save your code as **Q05cFINISHED** with the correct file extension for the programming language.

(8)

(Total for Question 5 = 14 marks)

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1. Question 5 · part c · 8 marks · Working and answer

Working and answer

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2. Question 3 · part a · 6 marks

2019 JUN · Paper 02

Further practice · Attempt part a only.

3 Different algorithms can be used to manipulate data.

(a) **Email.txt** contains a list of email addresses.

Open **Q03a** in the code editor.

Write a program to implement these requirements.

The code must:

- check each email address to ensure it contains the '@' symbol.
- write email addresses that do not contain the '@' symbol to an **Error.txt** file.

You must use the structure given in the file **Q03a** to complete the program.

Do not add further functionality.

Save your code as **Q03aFINISHED** with the correct file extension for the programming language.

(6)

(b) Explain **one** drawback of using the merge sort algorithm to sort large data sets.

(2)

(c) Open **Q03c** in the code editor.

Write a program to display the square and cube of a number between 1 and 50 entered by a user.

The code must:

- ask the user to enter a number between 1 and 50 inclusive
- display the number, the square of the number and the cube of the number, with appropriate labels
- stop when a number outside the range 1 to 50 is entered.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(6)

(Total for Question 3 = 14 marks)

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2. Question 3 · part a · 6 marks · Working and answer

Working and answer

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3. Question 3 · part a · 8 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

3 Several encryption algorithms have been developed.

(a) Ahmed has started to create a Caesar cipher encryption program.

Caesar cipher encryption works by giving a number value to a key.

Each letter in a plaintext message is replaced by a new ciphertext letter using the key.

For example, as shown in the table, a key value of +2 would change the plaintext message **encrypt** to the ciphertext message **gpetarv**.

Plaintext	e	n	c	r	y	p	t
Ciphertext	g	p	e	t	a	r	v

Ahmed's program already converts from plaintext to ciphertext.

He wants to extend the program to:

- accept the input of a plaintext message
- accept the input of the key, which must be a positive integer between 1 and 25
- write the generated ciphertext to a text file named **Cipher.txt**
- display the ciphertext.

Open **Q03a** in the code editor.

Complete the program code.

You must use the structure given in the file **Q03a** to complete the program.

Do not add any further functionality.

Save your code as **Q03aFINISHED** with the correct file extension for the programming language.

(8)

(b) Explain **one** reason why the Caesar cipher is a weak encryption algorithm.

(2)

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3. Question 3 · part a · 8 marks · Working and answer

Working and answer

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4. Question 5 · part a · 9 marks

2022 JUN · Paper 02

Further practice · Attempt part a only.

5 Julia runs a computer gaming club.

(a) She wants a program to check passwords stored in a file.

The file **passwords.txt** contains the list of passwords.

The program must:

- check each password to ensure that:
  - the first character is an uppercase letter
  - if it is, that it also includes at least one digit (0–9)
- if a password does not meet these requirements:
  - display the password
  - increment the number of incorrect passwords
- display the total number of incorrect passwords after all the passwords have been checked.

Open **Q05a** in the code editor.

Write the program.

You must use the structure given in **Q05a** to write the program.

Do not add any further functionality.

Save your code as **Q05aFINISHED** with the correct file extension for the programming language.

(9)

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4. Question 5 · part a · 9 marks

2022 JUN · Paper 02

Further practice · Attempt part a only.

(b) **Figure 3** shows an array that stores player scores after a game.

3	2	10	8	1	9

Figure 3

(i) Julia uses a bubble sort algorithm to sort the scores.

Complete the table to show how the bubble sort algorithm will sort the scores.

You may not need to use all the rows.

(3)

3	2	10	8	1	9

(ii) Explain **one** reason why a bubble sort is very efficient in terms of memory usage.

(2)

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4. Question 5 · part a · 9 marks · Working and answer

Working and answer

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5. Question 5 · part c · 6 marks

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Further practice · Attempt part c only.

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5 A marine scientist is conducting a study of tuna in the world's oceans.

(a) A list of tuna species needs to be sorted into **descending** alphabetical order using a merge sort algorithm.

Figure 5 shows the lists created at the end of the splitting process.

Each list is a single tuna species.

Bigeye	Blackfin	Albacore	Longtail	Bluefin

Figure 5

It will require three passes to merge the lists into a single sorted list in **descending** order.

Complete the merge sort using the space provided.

(2)

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5. Question 5 · part c · 6 marks

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Further practice · Attempt part c only.

(b) A bubble sort could be used to sort the list of tuna species into **ascending** order.

Figure 6 shows an algorithm for a bubble sort.

```
1  SET myTuna TO ["Bigeye", "Blackfin", "Albacore",
2                 "Longtail", "Bluefin"]
3  SET tmp TO 0
4  SET swaps TO True
5  SET length TO LENGTH (myTuna)
6
7  WHILE (swaps = True) DO
8      SET swaps TO False
9      FOR ndx FROM 0 TO length - 1 DO
10         IF myTuna[ndx] > myTuna[ndx + 1] THEN
11             SET tmp TO myTuna[ndx + 1]
12             SET myTuna[ndx] TO myTuna[ndx + 1]
13             SET myTuna[ndx + 1] TO tmp
14             SET swaps TO True
15         END IF
16     END FOR
17 END WHILE
```

Figure 6

There is an error in the loop between line 9 and line 16.

Complete the table to give the line number with an error and a corrected line of pseudocode.

(2)

Line number with error	
Corrected line of pseudocode	

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P 7 2 9 3 8 A 0 1 2 1 6

5. Question 5 · part c · 6 marks

2023 JUN · Paper 02

Further practice · Attempt part c only.

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(c) The scientist is collecting and storing data about tuna.

The data collected is:

- species
- length in centimetres
- weight in kilograms
- age in years.

The data is stored in an array.

The collected data needs to be written to a file named **TunaData.txt**

Each record stored in the file must have a code number in the first field.

Code numbers must start at 101

Each field in a record should be separated by a comma.

Figure 7 shows the contents of the file.

```
101,Yellowfin,105,15,3
102,Albacore,90,15,5
103,Skipjack,50,3,4
104,Bigeye,105,25,4
105,Atlantic Bonito,50,4,2
106,Northern Bluefin,190,120,11
107,Southern Bluefin,190,120,11
108,Tongol,90,20,4
```

Figure 7

Open **Q05c** in the code editor.

Write the program.

You must use the structure given in **Q05c** to write the program.

Do not add any further functionality.

Save your code as **Q05cFINISHED** with the correct file extension for the programming language.

(6)

(Total for Question 5 = 10 marks)

5. Question 5 · part c · 6 marks · Working and answer

Working and answer

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6. Question 5 · part b · 8 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

5 Martha is a computer science student and is learning to code.

(a) **Figure 7** shows an algorithm in pseudocode written by Martha.

```
1  SET first TO 0
2  SET second TO 0
3  SET answer TO 0
4  SET user_answer TO 0
5  SET score TO 0
6
7  REPEAT 10 TIMES
8      SET score TO 0
9      SET first TO RANDOM(99)
10     SET second TO RANDOM(99)
11     SET answer TO first + second
12
13     SEND first "+" second "=" TO DISPLAY
14     RECEIVE user_answer FROM (INTEGER) KEYBOARD
15
16     IF user_answer = answer THEN
17        SET score TO score + 1
18     END IF
19 END REPEAT
20 SEND score TO DISPLAY
```

Figure 7

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6. Question 5 · part b · 8 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

There is a logic error on line 8 of the pseudocode.

(i) Explain the problem this error will cause.

(2)

(ii) Describe the purpose of the algorithm shown in **Figure 7**, assuming the error has been corrected.

(2)

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6. Question 5 · part b · 8 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

(b) Martha is writing a program to calculate grades.

Each student takes three tests. Each test has a maximum score of 50.

Scores are stored in a two-dimensional array called tbl_scores.

The lowest of the three scores is ignored and the other two are added to get a total out of 100.

Grades are based on the table in **Figure 8**.

Total	Grade
80 or more	Distinction
65 to 79	Merit
50 to 64	Pass
49 or less	Fail

Figure 8

Each student's total and grade are saved to a text file.

For example, the first student scores 45, 42 and 43. The lowest score 42 is ignored. The total is 88 and the grade is Distinction.

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6. Question 5 · part b · 8 marks

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Further practice · Attempt part b only.

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Figure 9 shows part of the text file.

```
88  Distinction
77  Merit
96  Distinction
67  Merit
48  Fail
76  Merit
```

Figure 9

The program must:

- calculate the total and grade for each student
- write total and grade to a new line in the file **grades.txt**
- display a message to say the grades file has been created.

Open **Q05b** in the code editor.

Write the program.

You must use the array given in **Q05b**.

Do not add any further functionality.

Save your code as **Q05bFINISHED** with the correct file extension for the programming language.

(8)

(Total for Question 5 = 12 marks)

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6. Question 5 · part b · 8 marks · Working and answer

Working and answer

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